

Advanced Econometrics – Exam

September 2, 2024

Exercise 1 (40%)

This exercise is based on an example from the textbook by Wooldridge. "The general fertility rate (*gfr*) is the number of children born to every 1,000 women of childbearing age." Using the *fertil3* dataset from the *wooldridge* package, an economist estimated some models for *gfr*. The data covered the years 1913 to 1984.

The first specification considered by the economist was

$$gfr_t = \beta_0 + \beta_1 pe_t + \beta_2 ww2_t + \beta_3 pill_t + \varepsilon_t, \quad (1)$$

"explains *gfr* in terms of the average real dollar value of the personal tax exemption (*pe*) and two binary variables. The variable *ww2* takes on the value unity during the years 1941 to 1945, when the United States was involved in World War II. The variable *pill* is unity from 1963 onward, when the birth control pill was made available for contraception."

Subsequently, the economist noted that "the fertility rate may react to changes in *pe* with a lag". Therefore, the economist estimated a model with the following specification

$$gfr_t = \beta_0 + \beta_1 pe_t + \beta_2 pe_{t-1} + \beta_3 pe_{t-2} + \beta_4 ww2_t + \beta_5 pill_t + \varepsilon_t. \quad (2)$$

Considering this specification, the economist observed that "there is substantial correlation between pe_t , pe_{t-1} , and pe_{t-2} , and this multicollinearity makes it difficult to estimate the effect at each lag." Additionally, the economist performed the F test to check whether pe_t , pe_{t-1} , and pe_{t-2} are jointly significant.

Finally, the economist estimated two additional models with the following specifications.

$$gfr_t = \beta_0 + \beta_1 pe_{t-2} + \beta_2 ww2_t + \beta_3 pill_t + \varepsilon_t \quad (3)$$

$$gfr_t = \beta_0 + \beta_1 pe_{t-3} + \beta_2 ww2_t + \beta_3 pill_t + \varepsilon_t \quad (4)$$

$$gfr_t = \beta_0 + \beta_1 ww2_t + \beta_2 pill_t + \varepsilon_t \quad (5)$$

The quality publication table on the next page shows estimates of the abovementioned models. The output below contains the F test results and stationarity analysis of *gfr* and its first differences.

Use $\alpha = 5\%$ **significance level** while answering the following tasks.

- (5%) How could the economist explain the fact that *gfr* reacts with a lag on changes in *pe*? Explain.
- (5%) Considering specifications (1)-(5), which presents DL models and which presents ARDL models? Explain.
- (5%) Which model should be selected from the quality publication table as superior? Explain.
- (10%) Calculate and interpret the long-term multiplier for model (2).
- (5%) Interpret the estimate for pe_t in model (2) in the quality publication table.
- (10%) Discuss the validity of the estimates presented in the quality publication table in the light of the stationarity test results for *gfr*. Explain.

Table 1: Quality Publication Table with estimates of models (1)-(5)

	<i>Dependent variable: gfr</i>				
	<i>OLS</i>	<i>dynamic linear</i>			
	(1)	(2)	(3)	(4)	(5)
pe	0.083*** (0.030)	0.073 (0.126)			
L(pe)		-0.006 (0.156)	0.086*** (0.028)		
L(pe, 2)		0.034 (0.126)		0.093*** (0.027)	
ww2	-24.238*** (7.458)	-22.126** (10.732)	-20.716*** (6.991)	-16.472** (6.669)	-16.511** (7.253)
pill	-31.594*** (4.081)	-31.305*** (3.982)	-31.403*** (4.024)	-31.256*** (3.934)	-27.597*** (4.003)
Constant	98.682*** (3.208)	95.870*** (3.282)	97.795*** (3.159)	96.438*** (3.138)	105.211*** (2.294)
Observations	72	70	71	70	72
R ²	0.473	0.499	0.481	0.495	0.413
Adjusted R ²	0.450	0.459	0.458	0.472	0.396
Residual Std. Error	14.685	14.270	14.457	14.104	15.387
F Statistic	20.378***	12.728***	20.713***	21.558***	24.312***

Note: *p<0.1; **p<0.05; ***p<0.01; L(pe)= pe_{t-1} ; L(pe,2)= pe_{t-2}

```
> linearHypothesis(m2, c("pe=0", "L(pe)=0", "L(pe, 2)=0"))
Linear hypothesis test
```

Hypothesis:

pe = 0

L(pe) = 0

L(pe, 2) = 0

Model 1: restricted model

Model 2: $gfr \sim pe + L(pe) + L(pe, 2) + ww2 + pill$

```
Res.Df  RSS Df Sum of Sq    F Pr(>F)
1      67 15460
2      64 13033  3    2427.1 3.973 0.01165 *
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

> testdf2(variable = fertil3$gfr, # vector tested
+         test.type = "nc", # test.type = "nc",
+         max.augmentations = 5, # maximum number of augmentations added
+         max.order=5) # maximum order of residual lags for BG
  augmentations      adf      p_adf bgodfrey.statistic bgodfrey.statistic.1
1              0 -1.7961918 0.07261885              71          5.87103860
2              1 -1.5050019 0.13338838              70          0.23835379
3              2 -1.5998236 0.10323599              69          0.19980527
4              3 -1.3010202 0.19825258              68          0.16435302
5              4 -1.0881310 0.26594929              67          0.08218637
6              5 -0.9809652 0.30002695              66          0.06302808
  bgodfrey.statistic.2 bgodfrey.statistic.3 bgodfrey.statistic.4
1          8.2930760          12.9712374          14.8134527
2          4.7857989          6.4814911          8.2701476
3          0.3634352          3.9875888          6.9239462
4          0.1994200          0.3038063          1.0304069
5          0.6675841          0.6957293          0.7460225
6          0.1828108          0.1828423          0.2157320
  bgodfrey.statistic.5 p_bg.p.value.1 p_bg.p.value.2 p_bg.p.value.3
1          16.7790218          0.01539197          0.01581909          0.004699221
2          11.2134741          0.62539763          0.09136439          0.090395313
3           9.7255767          0.65487808          0.83383679          0.262807436
4           4.9566360          0.68518001          0.90509987          0.959311043
5           2.4652293          0.77435609          0.71620269          0.874208001
6           0.2677784          0.80177240          0.91264765          0.980310418
  p_bg.p.value.4 p_bg.p.value.5
1          0.005104186          0.004938528
2          0.082168802          0.047307904
3          0.139962322          0.083394980
4          0.905150792          0.421195308
5          0.945534470          0.781722267
6          0.994584362          0.998205549

```

```
kpss.test.d <- ur.kpss(diff.xts(fertil3$gfr),
                        type = c("mu")) # constant deterministic component
> summary(kpss.test.d)
```

```
#####
# KPSS Unit Root Test #
#####
```

Test is of type: mu with 3 lags.

Value of test-statistic is: 0.1819

Critical value for a significance level of:

	10pct	5pct	2.5pct	1pct
critical values	0.347	0.463	0.574	0.739

```
> pp.test <- ur.pp(diff.xts(fertil3$gfr),          # tested series
+                  type = c("Z-tau"),             # standardization of the test statistic
+                  model = c("constant"))          # constant deterministic component
> # which means we assume that any trends in the data are stochastic
>
> summary(pp.test)
```

```
#####
# Phillips-Perron Unit Root Test #
#####
```

Value of test-statistic, type: Z-tau is: -6.0688

	aux. Z statistics
Z-tau-mu	-1.242

Critical values for Z statistics:

	1pct	5pct	10pct
critical values	-3.525169	-2.90292	-2.588557

Exercise 2 (30%)

A different economist was intrigued by The Washington Post's fun article about the Hot Dog Eating Contest in Coney Island¹. She decided to analyse the factors that affect how many hot dogs a competitor can eat during 10 minutes contest. She downloaded the data from the last edition of the contest. The dependent variable `hotdogs` measured the number of hot dogs (count). As regressors she decided to use: `ever_champion` - a dummy variable (=1 if a person won any contest of this sort in the past), `BMI` - Body Mass Index (continuous), `neck` - length of the competitor's neck (in centimetres), and `exper` - experience measured of years of competing in contests of this sort. The main interest of the economist was to verify a hypothesis from the Washington Post that a shorter neck helps to eat more hot dogs.

Use $\alpha = 5\%$ significance level.

- (5%) Interpret the parameters of the model.
- (5%) How to verify the hypothesis regarding the neck length? Explain.
- (5%) Which other models could be used to model the number of hot dogs eaten in a contest? Explain.
- (5%) What other variable could be used to improve the model? Explain.
- (10%) The economist estimated another model with neck squared. The original model had the likelihood value equal -213.1799, while the new one had the loglikelihood equal -212.5207. Which model should be selected as superior to the other? Explain.

Upper-tail critical values of the chi-square distribution

degrees of freedom	Probability less than the critical value				
	0.90	0.95	0.975	0.99	0.999
1	2.706	3.841	5.024	6.635	10.828
2	4.605	5.991	7.378	9.210	13.816
3	6.251	7.815	9.348	11.345	16.266
4	7.779	9.488	11.143	13.277	18.467
5	9.236	11.070	12.833	15.086	20.515
6	10.645	12.592	14.449	16.812	22.458

¹<https://www.washingtonpost.com/sports/2023/07/01/hot-dog-eating-contest-competitive-eater/>


```
> pois = glm(y~ever_champion+bmi+neck+exper, family = poisson)
> summary(pois)
```

Call:

```
glm(formula = y ~ ever_champion + bmi + neck + exper, family = poisson)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-5.466	-2.588	-1.093	1.014	11.810

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.973827	0.067616	14.402	< 2e-16	***
ever_champion	0.146432	0.023780	6.158	7.37e-10	***
bmi	0.191906	0.012388	15.492	< 2e-16	***
neck	-0.055625	0.002997	-18.558	< 2e-16	***
exper	0.093222	0.004146	22.484	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 8836.3 on 999 degrees of freedom
Residual deviance: 7766.1 on 995 degrees of freedom
AIC: 10687

Exercise 3 (30%)

Kudła et al. (2023) estimated ordered logit models for inheritance and gift taxation among the EU countries. They "analyzed the impact of a series of factors, such as the country's affluence, political preferences, preferences for equity, ageing ratio, fiscal standing of the state, and the country's size, on inheritance tax systems". One of the research hypotheses was:

"Hypothesis 4: High inheritance and gift taxation depend positively on the proportion of elderly people to young people in a society."

Another research hypothesis was:

"Hypothesis 5: In highly populated [Big] countries the taxation of inheritance and gifts is higher."

The dependent variable quantified different types of tax designs. The "levels were coded as cardinal numbers from 0 (no tax) to 3 (progressive tax with exemptions) and used as the dependent variable".

"The data about the citizens' wealth were taken from the Credit Suisse Global Wealth Databook. The political variables were obtained from the V-Dem Dataset, version 11.1 (Varieties of Democracy) [Lührmann et al., 2020; Pemstein et al., 2020]. The other variables, including demographic, economic, and financial indicators, were gathered from Eurostat."

"The Independence of youth is a percentage of the population aged between 18 years and 34 years and living with parents. The variable measures the need for the attention of parents (which should be lower for a higher share of children living together with parents) and the possible attractiveness of the financial support delivered by parents to their children. The cost of attention should be lower when young people live together with their parents and conversely, higher independence requires higher gift or inheritance promises. A high share in the population of persons age 75 or higher (75+ share) is a proxy for the potential number of wealth transfers transferred to young individuals. The Ageing ratio is the proportion of people aged greater than 64 years to those aged less than 21 years. The greater value of this ratio indicates a population ageing and a decrease in the bargaining power of parents (the attention is less available and with higher cost because the number of young people is insufficient to provide attention to older people). The perspective of inheritance should also be affected by life expectancy, which is approximated by the expected number of years of future life at birth (Life expectancy). Finally, we add the sign of net migration, Migration (sign). If immigration was higher than emigration, then this variable took the value 1, and otherwise it took 0. We use the sign of migration instead of immigration and emigration difference to obtain convergence in the estimation of parameters. The net inflow of people should alleviate the problem of the limited attention provided by the younger population to their parents as it increases the number of people who can provide care to older people. It simultaneously decreases the meaning of inheritances and gifts, which alleviates the need for complicated and burdensome inheritance taxation. The second group of variables refers to the wealth of a country and includes gross domestic product per capita (GDP per capita), gross domestic product per capita in purchasing power standard (GDP per capita in PPS), and wealth per adult (Wealth per adult). The first two variables approximate the affluence of the country and the third refers to the accumulated wealth of a representative citizen. They can be interrelated, and so we apply them in separate econometric models. The third group of variables consists of some macroeconomic variables, like Inflation measuring the consumer price index."

"The policy orientation of the state is captured by three variables (Varieties of Democracy data [V-Dem Dataset, version 11.1]):

- orientation of the ruling party (Economic left-right scale), defined on a scale from 0 to 5. The scale starts at far left (0), through left (1), center left (2), center (3), center right (4), and right (5), to far right (6), where left means a higher preference for the active role of government in the economy; and
- (3) the illiberalism index (Illiberalism) taken from V-Dem, which measures the "extent to which the party shows a lack of commitment to democratic norms prior to elections." It represents the measure of populism in a society. The variable takes values in the range 0–1."

"Finally, we add a dummy variable (Big countries) to distinguish between the more populated countries and the other ones. If the country in question is France, Germany, Italy, Poland, Spain, or the United Kingdom,

this variable takes on a value of 1, and it takes 0 otherwise. The larger countries easier enforce the payment of inheritance taxes and can benefit more from large inheritances (high accumulated wealth)."

Use $\alpha = 5\%$ significance level.

- a) (5%) Was the ordered logit model a good choice for the dependent variable? Explain
- b) (5%) List all significant variables in model (1).
- c) (5%) Which model should be selected as superior out of those presented? Explain.
- d) (5%) How to verify Hypothesis 4? Explain.
- e) (5%) How to verify Hypothesis 5? Explain.
- f) (5%) List diagnostic tests that can be used for ordered choice models.

Dependent: Tax design type	(1)	(2)	(3)	(4)
Independence of youth	0.05 (0.05)	0.06 (0.05)	0.05 (0.05)	0.04 (0.05)
75+ share	281.55*** (83.62)	284.99*** (79.81)	279.08*** (79.52)	248.07*** (76.30)
Aging ratio	-15.79*** (4.96)	-17.39*** (4.90)	-15.82*** (4.70)	-16.88*** (4.77)
Life expectancy	-0.61* (0.34)	-0.86** (0.36)	-0.56* (0.31)	-0.04 (0.22)
Migration (sign)	-0.37 (0.86)	-0.36 (0.87)	-0.37 (0.87)	-0.28 0.59
GDP per capita	13.52 (59.00)			
GDP per capita PPS		15,485.99 (13,033.01)		
Wealth per adult			-0.002 (0.088)	
Inflation	-0.15 (0.14)	-0.15 (0.14)	-0.15 (0.14)	
Economic left-right scale	0.41 (0.28)	0.40 (0.28)	0.41 (0.28)	0.34 (0.27)
Illiberalism	-1.80 (2.19)	-1.39 (2.22)	-1.82 (2.19)	-1.17 (1.97)
Big countries [#]	15.82*** (3.93)	17.21*** (4.08)	16.23*** (3.92)	
Countries	27	27	27	27
Joint insignificance (chi-square statistics)	36.74**	37.43**	36.81**	21.63***
Std. dev. of the individual effect	13.29	13.73	13.52	13.62
Individual effects insignificance	380.06***	407.51***	400.03***	463.74***
AIC	403.6	417.5	418.7	407.8
BIC	460.2	510.5	511.7	460.4

[#]Big countries include France, Germany, Italy, Poland, Spain and the United Kingdom.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

GDP, gross domestic product; PPS, purchasing power standard, AIC, Akaike Information Criterion, BIC, Bayesian Information Criterion.

